

DecodeLabs — Project 1: Data Cleaning & Preparation

Data Quality Change Log & Accountability Report

This document records every transformation applied to the raw dataset (**Dataset_for_Data_Analytics.xlsx**) during cleaning, along with the business impact of each change, per the Phase 4 documentation standard ("Stakeholders need to know WHAT you changed and WHY").

Change ID	Description	Impact	Status
CR000	Loaded raw dataset	1200 rows, 14 columns	Resolved
CR001	Imputed missing 'CouponCode' values with explicit 'NONE' category (no coupon applied) instead of deleting rows	Preserved 309 records that would otherwise have been lost	Resolved
CR002	Removed fully duplicated rows (identical across all columns)	Removed 0 duplicate rows	Resolved
CR003	Enforced uniqueness on 'OrderID' (kept first occurrence of any repeats)	Removed 0 duplicate Order ID(s). Verified: 0% duplicate-ID error rate	Resolved
CR004	Standardized 'Date' column to ISO 8601 format (YYYY-MM-DD)	Converted all rows; 0 unparseable date(s) flagged. Verified: 0% date-format error rate	Resolved
CR-CASE-Product	Standardized text casing in 'Product' to Title Case	0 value(s) corrected	Resolved
CR-CASE-PaymentMethod	Standardized text casing in 'PaymentMethod' to Title Case	0 value(s) corrected	Resolved
CR-CASE-OrderStatus	Standardized text casing in 'OrderStatus' to Title Case	0 value(s) corrected	Resolved
CR-CASE-ReferralSource	Standardized text casing in 'ReferralSource' to Title Case	0 value(s) corrected	Resolved
CR005	Enforced 2-decimal numeric precision on 'UnitPrice' and 'TotalPrice'	Ensures consistent currency formatting across all records	Resolved
CR006	Validated 'TotalPrice' = Quantity x UnitPrice for every row and recalculated any mismatches	0 row(s) corrected	Resolved
CR007	Checked for non-positive Quantity/UnitPrice values	0 suspicious row(s) found (flagged, none removed automatically)	Resolved
CR008	Final verification gate	Rows 1200->1200. 0% duplicate-ID rate, 0% bad-date rate, 0 nulls remaining.	Resolved

Final Verification Gate

0% error rate on unique identifiers (OrderID). 0% error rate on date formats (ISO 8601). 0 remaining null values. Dataset approved as production-ready "gold standard" data, cleared to proceed to Project 2.